

Advanced (Habanero)

- Q1.** What is the graph of the inequality $x \geq 2$?
- (A) A vertical line at $x = 2$
(B) A horizontal line at $y = 2$
(C) A solid line at $x = 2$ and shading to the right
(D) A dashed line at $x = 2$ and shading to the left
(E) A solid line at $x = 2$ and shading to the left
- Q2.** Which inequality has the solution region below the line $y = x - 1$?
- (A) $y \leq x - 1$
(B) $y \geq x - 1$
(C) $y < x - 1$
(D) $y > x - 1$
(E) $x \leq y + 1$
- Q3.** What type of line is used for the inequality $x + y \leq 3$?
- (A) Solid line
(B) Dashed line
(C) Dotted line
(D) Horizontal line
(E) Vertical line
- Q4.** The inequality $y > x$ has a boundary line that is
- (A) Solid and vertical
(B) Dashed and diagonal
(C) Solid and horizontal
(D) Dashed and horizontal
(E) Solid and diagonal
- Q5.** The solution region for the inequality $x \geq 0$ and $y \geq 0$ is
- (A) The first quadrant
(B) The second quadrant
(C) The third quadrant
(D) The fourth quadrant
(E) The entire coordinate plane
- Q6.** The inequality $y \leq -x$ has a solution region that is
- (A) Above the line $y = -x$
(B) Below the line $y = -x$
(C) To the left of the line $y = -x$
(D) To the right of the line $y = -x$
(E) On the line $y = -x$
- Q7.** Which of the following is the graph of the inequality $y < 2x - 3$?
- (A) A dashed line at $y = 2x - 3$ and shading below
(B) A solid line at $y = 2x - 3$ and shading above
(C) A dashed line at $y = 2x - 3$ and shading above
(D) A solid line at $y = 2x - 3$ and shading below
(E) A vertical line at $x = 3$ and shading to the left
- Q8.** The inequality $x + 2y \geq 4$ has a boundary line with a slope of
- (A) $-\frac{1}{2}$
(B) $\frac{1}{2}$
(C) 2
(D) -2
(E) 0

Open Ended Questions (FRQ)

- Q9.** Graph the inequality $y > x - 2$ and identify the solution region.
- Q10.** Graph the system of inequalities $x \geq 0, y \geq 0, x + y \leq 4$ and identify the solution region.

Chef's Tips

- Tip 1: Encourage students to use graph paper.
- Tip 2: Remind students to label their axes.
- Tip 3: Emphasize the importance of shading the correct region.